

YEAR 10 SCIENCE (MODIFIED)
CHEMICAL SCIENCE
TEST 2

NAME: SOLUTIONSMARK: 47

Analyses how the periodic table organises elements and use it to make predictions about the properties of elements.

(Mult. choice Q 1 - 6, Part 2 Q 1 - 2)

12

Explains how chemical reactions are used to produce particular products and how different factors influence the rate of reactions.

(Mult. choice Q 7 - 13, Part 2 Q 3 - 7)

35**Part 1: Multiple Choice**

Write your choice for each question in the table below.

1. How many **protons** does a sodium atom have?

- (a) 12
 (b) 10
 (c) 11
 (d) 5

2. The **atomic number** of phosphorus is:

- (a) 15
 (b) 16
 (c) 31
 (d) 30

3. The charge of an electron is:

- (a) negative.
 (b) neutral.
 (c) positive.
 (d) nothing.

4. The **atomic number** of an element represents:

- (a) the number of electrons.
 (b) the number of protons.
 (c) the number of neutrons.
 (d) the number of protons added to number of electrons.

5. The nucleus of an atom consists of:

- (a) electrons and protons.
 (b) electrons and neutrons.
 (c) protons only.
 (d) protons and neutrons.

QUESTION	ANSWER
1	C
2	A
3	A
4	B
5	D
6	C
7	C
8	A
9	B
10	B
11	C
12	D
13	C

6. How many *protons* does a sodium atom have?
- (a) 12
 - (b) 10
 - (c) 11
 - (d) 5
7. The formula for dinitrogen tetrahydride is
- (a) N_2H_5 .
 - (b) N_4H_2 .
 - (c) N_2H_4 .
 - (d) N_3H_2 .
8. Which of the following tests would indicate that a colourless, odourless gas is *hydrogen*?
- (a) A burning match placed in the gas produces a 'pop' sound.
 - (b) A burning match placed in the gas flares up.
 - (c) A sample of the gas turns limewater milky.
 - (d) A burning match placed in the gas goes out.
9. Which of the following is likely to contain *ions* bonded together?
- (a) CO
 - (b) Na_2SO_4
 - (c) NO_2
 - (d) F_2
10. What name would we give to the substance KCl ?
- (a) potassium monochloride
 - (b) potassium chloride
 - (c) potassium chlorine
 - (d) monopotassium chloride
11. Which of the following is a *base*?
- (a) copper sulphate
 - (b) ammonium nitrate
 - (c) sodium hydroxide
 - (d) sodium chloride
12. Which of the following is the correct general equation to describe the reaction of an acid and a base?
- (a) acid + base $\rightarrow$ hydrogen + water
 - (b) acid + base $\rightarrow$ carbon dioxide + metal hydroxide
 - (c) acid + base $\rightarrow$ carbon dioxide + water
 - (d) acid + base $\rightarrow$ salt + water
13. Which is the correct formula for *nitric acid*?
- (a) H_2SO_4
 - (b) CH_3COOH
 - (c) HNO_3
 - (d) H_3PO_4

Short Answer

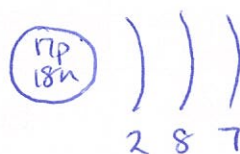
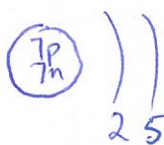
Write the answer to the questions in the spaces provided.

1. Complete the following table carefully.

ELEMENT	NUMBER OF PROTONS	NUMBER OF NEUTRONS	NUMBER OF ELECTRONS
$^{39}_{19}\text{K}$	19	20	19
^7_3Li	3	4	3
$^{19}_9\text{F}$	9	10	9

(3)

2. Draw diagrams to show the electron configurations of the following elements. **Remember to include the particles in the nucleus.**



(Nucleus - 1 mark Shells - 1 mark)

(4)

3. (a) When naming some compounds (called covalent compounds), prefixes are used. Write the number that corresponds with the following prefixes. **(Note: They are not in order.)**

tetra 4 hexa 6 di 2 penta 5

(2)

($\frac{1}{2}$ mark each)

- (b) Complete the table below by writing the name of the compounds shown. **(You will need the list of ions to help.)**

COMPOUND	FORMULAE
K_2CO_3	potassium carbonate
CuSO_4	copper sulphate
NaCl	sodium chloride
$\text{Zn}(\text{NO}_3)_2$	zinc nitrate
N_2O_4	dinitrogen tetroxide
$\text{Ba}(\text{OH})_2$	barium hydroxide
CO_2	carbon dioxide

(1 mark each)

(7)

4. Complete the following table by writing down **how many atoms of each element** are found in the compounds. **Make sure you name the elements.** The first one is done for you.

COMPOUND	NUMBER OF ELEMENTS
H ₂ O	2 x hydrogen, 1 x oxygen
CuCO ₃	1x copper, 1x carbon, 3x oxygen (2)
CH ₃ COOH	2x carbon, 4x hydrogen, 2x oxygen (2)
Mg(OH) ₂	1x magnesium, 2x oxygen, 2x hydrogen. (2)

(6)

5. Complete the following table for the acids and the salts they produce.

ACID	FORMULA	SALT PRODUCED
hydrochloric	HCl	chloride
nitric	HNO ₃	nitrate
sulphuric	H ₂ SO ₄	sulphate

($\frac{1}{2}$ mark each)

(3)

6. Complete the following general word equations for acid reactions.

ACID + METAL → SALT + HYDROGEN (1)

ACID + BASE → SALT + WATER (1)

ACID + CARBONATE → SALT + WATER
+ CARBON DIOXIDE (1)

(3)

7. Complete the following word equations. **Remember to name the salt correctly.**

